



A Course End Project Report on
Real-Time Wireless Notice Board using Bluetooth

A8421- Analog Communications Laboratory

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Abstract

Traditional notice boards require frequent manual updates, which are time-consuming and inefficient. This project presents a novel approach to modernizing notice board communication by implementing a Real-Time Wireless Notice Board using Bluetooth. The proposed system enables users to send text messages wirelessly from a mobile application to be displayed on a 16x2 LCD screen through an HC-05 Bluetooth module and an Arduino microcontroller. The system eliminates the need for physical updates, reducing effort and enhancing convenience.

This project aims to provide an accessible, cost-effective, and efficient solution for educational institutions, offices, and public places where information dissemination is crucial. By leveraging Bluetooth technology, the system offers a simple and reliable alternative to traditional notice boards. Additionally, this work explores potential future enhancements such as a Wi-Fi-enabled remote access system, allowing messages to be updated from anywhere over the internet, thus extending its usability and reach. The successful implementation of this system contributes to the ongoing shift toward digital transformation and sustainability by reducing paper usage and optimizing real-time communication.

Keywords: Wireless notice board, Bluetooth communication, Arduino, HC-05 module, real-time messaging, digital transformation.

1 Introduction

Traditional notice boards require manual updating, which is time-consuming and inefficient. This project addresses these limitations by implementing a wireless, real-time electronic notice board controlled via a mobile device. The integration of a Bluetooth communication system with an Arduino board ensures ease of operation, reliability, and affordability.

2 Literature Review

Several existing works have explored wireless notice boards using GSM, Wi-Fi, and RF modules. However, Bluetooth-based systems are preferred for their low power consumption, easy integration, and cost-effectiveness. Compared to GSM-based systems, which incur message costs, Bluetooth provides free local communication.

3 Objectives

- To develop a wireless electronic notice board controlled via Bluetooth.
- To enable real-time text display on a 16x2 LCD screen.
- To provide an efficient and cost-effective solution for notice board updates.
- To facilitate remote access through a mobile application.
- To explore enhancements using Wi-Fi connectivity for future scalability.

4 Methodology

4.1 Block Diagram

The system consists of:

- **Arduino Uno:** Microcontroller for processing and controlling the display.
- **HC-05 Bluetooth Module:** Enables wireless communication with a mobile device.
- **16x2 LCD Display:** Displays the received text message.
- **Mobile Application:** Sends messages via Bluetooth.

4.2 Circuit Diagram and Components

4.2.1 Components Used

- Arduino Uno
- HC-05 Bluetooth Module
- 16x2 LCD Display
- Potentiomet

The Figure 1 illustrates a Wireless Notice Board system using an Arduino microcontroller, an HC-05 Bluetooth module, and an LCD display. The system enables users to send messages wirelessly via a smartphone application, which are then displayed on the LCD screen. The Bluetooth module establishes a connection between the smartphone and Arduino, facilitating real-time message updates. This setup is useful for applications such as digital notice boards in educational institutions, offices, and public spaces.



Figure 1: Wireless Notice Board using Bluetooth

4.3 Working Principle

1. The mobile application connects to the HC-05 Bluetooth module.
2. Users send a text message via the mobile app.
3. The Bluetooth module transmits the message to Arduino.
4. The Arduino processes the data and displays it on the 16x2 LCD screen.

4.4 Software Implementation

- **Arduino IDE:** Used to program the Arduino microcontroller.
- **MIT App Inventor / Serial Bluetooth Terminal App:** Used to send text messages.

5 Results and Discussion

The prototype efficiently displays real-time text messages transmitted from a mobile device to the notice board. The system ensures minimal response time, typically less than one second, allowing for instantaneous updates without noticeable delays. Additionally, it operates with low power consumption, making it an energy-efficient solution for continuous use.

The system has demonstrated reliable performance in short-range communication, effectively transmitting messages within a range of approximately 10 meters. However, the reliance on Bluetooth communication imposes inherent range limitations. To overcome this constraint and enable long-distance communication, integrating a Wi-Fi-based enhancement would be a promising solution. This upgrade would allow remote access, enabling users to update the notice board from virtually anywhere within the network range, thereby significantly expanding the system's applicability in larger environments such as schools, offices, and public spaces.

6 Mapping POs and SDGs

6.1 Program Outcomes (POs) Mapping

Table 1: Mapping of Program Outcomes (POs) to Project Relevance

PO No.	Program Outcome	Relevance
PO1	Engineering Knowledge	Applied microcontroller and communication principles.
PO2	Problem Analysis	Identified inefficiencies in traditional notice boards.
PO3	Design/Development of Solutions	Designed a working prototype.
PO5	Modern Tool Usage	Utilized Arduino, HC-05, and mobile apps.
PO6	Engineer and Society	Provides a practical, user-friendly solution.

6.2 Sustainable Development Goals (SDGs) Mapping

Table 2: Mapping of Sustainable Development Goals (SDGs) to Project Relevance

SDG No.	Goal	Relevance
SDG 9	Industry, Innovation, and Infrastructure	Promotes smart infrastructure in educational institutions.
SDG 11	Sustainable Cities and Communities	Reduces paper wastage by replacing traditional notices.
SDG 12	Responsible Consumption and Production	Encourages digital transformation, reducing environmental impact.

7 Conclusion

This project successfully implements a Real-Time Wireless Notice Board using Bluetooth, eliminating manual updates. The system is low-cost, efficient, and user-friendly. Future improvements can enhance its range, security, and scalability.

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